

CHAPTER 7

Operations Planning and Control¹

7.1. Introduction

A large portion of modern industry consists of fabrication and assembly processes. Consider, for example, a plant that manufactures television sets, radios, and stereos. This plant must purchase materials, and perhaps parts, convert these materials into specific components, then assemble the components into the several products offered to the consumers. This greatly oversimplified process is represented in Figure 7.1.

Several observations are important at this point. First, there could be many different configurations of each of the three primary products. For example, there are many sizes and styles of television sets. Second, different products could contain several common components. It would not be surprising, for example, to find the same power switch in all three products. Third, different components could contain common raw materials. For example, certain components for each product may be stamped from the same roll of aluminum. The three components shown in Figure 7.2 could be for different products.

¹ A significant portion of this chapter has been adapted, with permission, from J.H. Mize, C.R. White, and G.H. Brooks, *Operations Planning and Control* (Englewood Cliffs, N.J.: Prentice Hall, 1971).

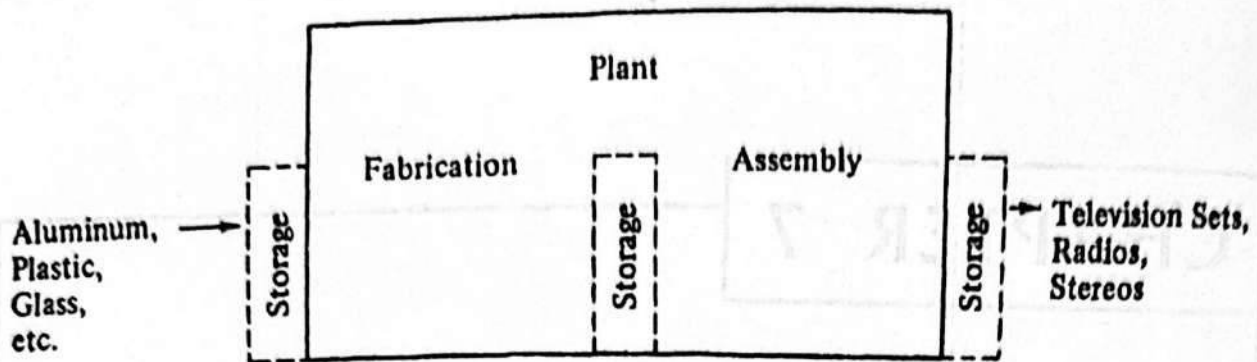


Fig. 7.1. Representation of a typical manufacturing plant.

7.2. Overview of Operations Planning and Control

Figure 7.3 is a schematic overview of an integrated operations planning and control system. The details of performance of the planning and control functions are presented in Sections 7.3 through 7.7. An intricate flow diagram, such as Figure 7.3, is conceptually difficult to understand in its entirety. Many of the component blocks shown in this master flow diagram can be expanded into detailed flow diagrams.

7.2.1 Demand Forecasting (I)

Inputs. The most tangible input to the demand forecasting function comes from product markets in the form of demand histories of existing products and, especially, trade information. Demand trends can be determined quantitatively both for one's own company and for the entire industry. Special trade information can either modify

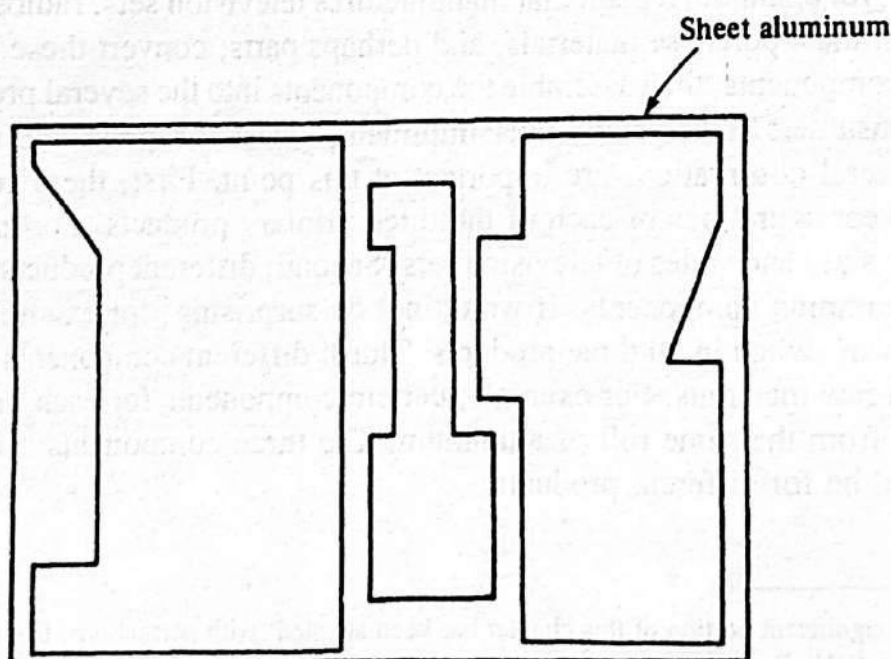
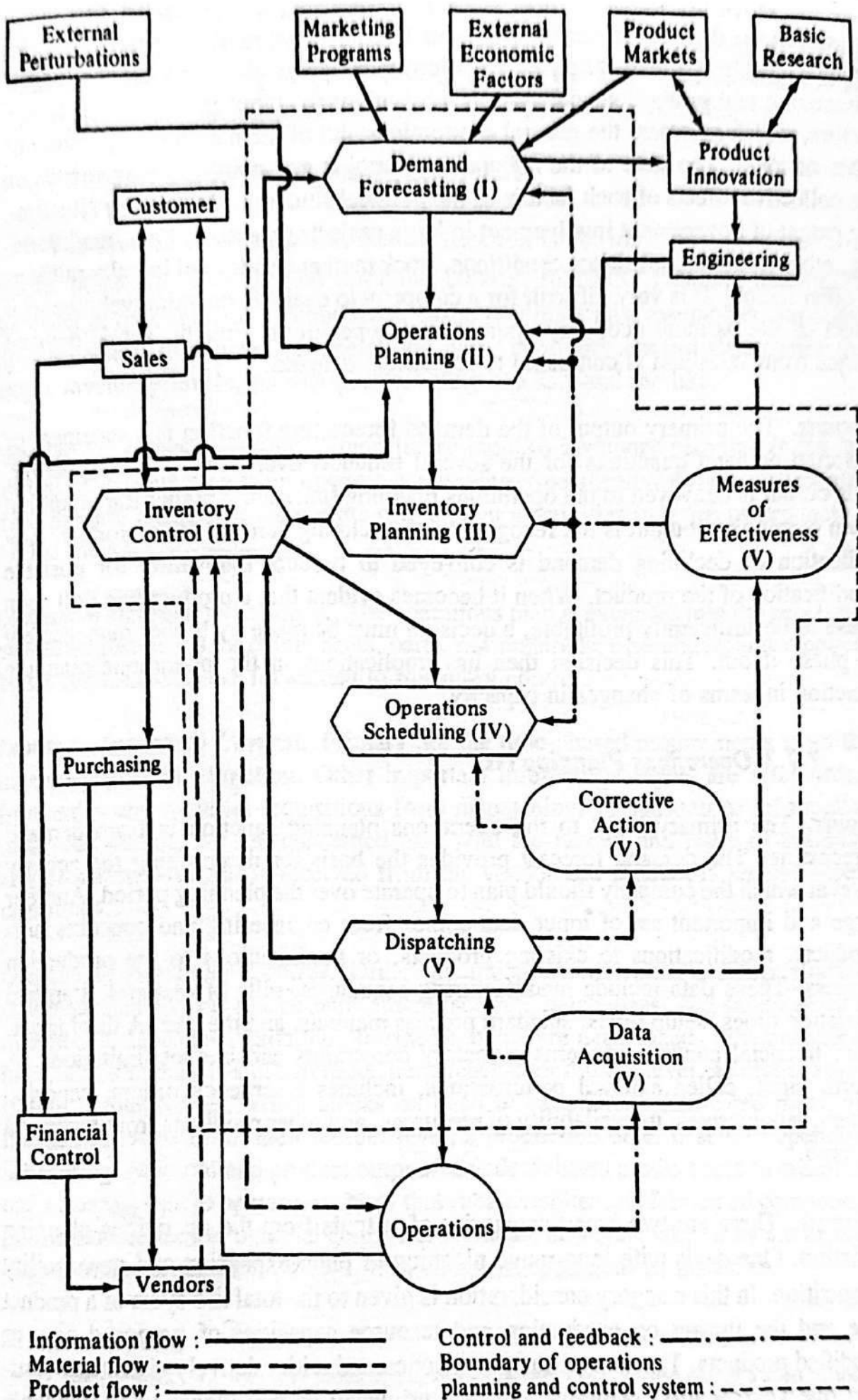


Fig. 7.2. Components for three different products stamped from the same sheet of aluminum.



or explain the demand trends. Another input comes from marketing programs. An attempt must be made to anticipate variations in the demand pattern that result from advertising and promotion schemes. A third important input is external economic factors, which represent the general economic health of the nation (and possibly of other nations). The state of the national economy at any point in time depends on the collective effects of such factors as the political situation, the military situation, the extent of government involvement in large projects (space, welfare, road building, other construction), labor conditions, stock market trends, and literally millions of other factors. It is very difficult for a company to evaluate quantitatively the total effect of the national economy upon its own operations. Finally, actual demand comes from sales and is compared to forecasted demand.

Outputs. The primary output of the demand forecasting function is a statement of expected demand quantities for the several products over some planning period. This output is conveyed to the operations planning function. Another important but often overlooked output is the recognition of declining demand for a product. Any indication of declining demand is conveyed to product innovation for possible modification of the product. When it becomes evident that a product line will soon cease to be sufficiently profitable, a decision must be made by higher management to phase it out. This decision then has implications in the operations planning function in terms of changes in capacity.

7.2.2 Operations Planning (II)

Inputs. The primary input to the operations planning function is from demand forecasting. The demand forecast provides the basis for determining the activity level at which the company should plan to operate over the planning period. Another large and important set of input data comes from engineering and concerns new products, modifications to existing products, or modifications to the production process. These data include manufacturing sequences, bills of material, standard operation times, setup times, standard process manuals, and the like. A third input, from financial control, concerns monetary constraints and budget limitations. A fourth input, called external perturbations, includes emergency orders, canceled orders, labor strikes, unavailability of resources, and other problems from the world at large.

Outputs. There are two broad categories of outputs from the operations planning function. One deals with long-range planning of plant expansion and new facility acquisition. In this category consideration is given to the total life cycle of a product line and the impact on production and resource capacities of proposed new or modified products. The second category, concerned with relatively short term (usually one year or less) planning, consists primarily of the allocation of available resources to production requirements. Machine groupings, facility layout, and as-

sembly line balancing are performed. Make or buy decisions are made, as are decisions regarding work force size, skill mix, staff leveling, number of shifts, approximate time phasing of production runs, and the like. This information is conveyed to inventory planning and control and operations scheduling.

7.2.3 Inventory Planning and Control (III)

The inventory planning and control function is shown in Figure 7.3 as two blocks: inventory planning and inventory control. Inventory planning determines the material requirements (components, parts, raw materials, assemblies, supplies, etc.) necessary to satisfy the operations plan. Inventory control determines the proper inventory levels, reorder points, safety stocks,² and the like.

Input to Inventory Planning. The primary input to inventory planning is the approved operations plan from operations planning. Specifically, the operations plan must provide a time-phased outline of the plant's activities in terms of production for the market.

Outputs of Inventory Planning. The operations plan is extended into the net, time-phased requirements for components, parts, raw materials, assemblies, and supplies. These requirements are forwarded to inventory control.

Inputs to Inventory Control. Primary are the time-phased requirements from the inventory planning function. Other important information inputs are sales orders from sales and material requisitions from dispatching. In addition to information input and output, we are concerned also with the receipt and issue of materials and products. Material is received from the vendor and product is received from operations.

Outputs of Inventory Control. Order quantities, reorder points, safety stock of raw materials and purchased parts, manufacturing batch sizes, and safety stock of assemblies, fabricated parts, and finished product are determined. When stocks on hand of raw materials and purchased parts reach their reorder level, a purchase order is sent to purchasing. When stocks on hand of assemblies, fabricated parts, and finished products reach their reorder level, a production order is sent to operations scheduling. Material and product outputs include finished product sent to customers and materials sent to operations. Note that subassemblies and fabricated components go from operations to material control as products; however, when they are brought back into the production process, they return to operations as materials.

²Safety stocks are needed as protection against unusually large demand during a short period of time.

7.2.4 Operations Scheduling (IV)

Inputs. Primary input to the scheduling function are requests for fabrication and assembly from inventory control. Another extremely important input comes from corrective action in the form of updated priorities and schedule adjustments of production orders currently in process.

Outputs. Detailed operation sequences for individual work activities, as well as start and stop times for all operations, are released to dispatching. New work is assigned to operating facilities, with proper consideration given to work already in process and to priority assignments. Schedule conflicts on production facilities are resolved. This function is at the heart of the entire planning and control system, since it is here that compromises must be made between economic batch sizes, due dates, resource constraints, manpower leveling, and facility utilization. Here also corrections are made for nonstandard process behavior.

7.2.5 Dispatching and Progress Control (V)

Since this function is shown in Figure 7.3 as four blocks, we discuss each block in turn.

Dispatching. This function is responsible for initializing production, that is, for releasing work orders to operations at the appropriate time. Material is requisitioned from materials control, and arrangements are made for changing the production process over to the new production item.

Data Acquisition. As work progresses, data are acquired from different processing points on the shop floor. Specific data required are the progress of work in production, the status of work awaiting processing, the status of production facilities, and the availability of required workers. Material usage is relayed back to inventory control. It should be noted that much of the data is also useful for accounting functions, quality control, personnel, etc.

Corrective Action. Short-term corrective action is performed here. Specific decisions are made regarding operating problems. These decisions include expediting critical jobs, determining new priorities, balancing workloads between work centers, handling personnel problems, problems with product quality, and equipment breakdowns, etc. These decisions are conveyed to operations scheduling and are included in the next execution of the scheduling function.

Measures of Effectiveness. Long-term corrective action is performed here. Control parameters are maintained to which measures of plant performance are related. Actual performance is compared with planned performance for such measures of effectiveness as production output, standard and premium labor cost, investment in

inventory, hiring and layoff cost, amount of scrap, facility utilization, schedule slippage, and so on. Particular attention is given to the identification of cause-effect relationships so that we can anticipate problems and try to prevent them. Information from this function constitutes the principal form of long-range system feedback and is used for recalculation of pertinent governing parameters.

7.2.6 Interfaces

The purposes of several of the interfaces shown in Figure 7.3 were discussed in the preceding sections; others are self-explanatory. Explained below are certain important interfaces whose functions are important to the operations planning and control system but whose purposes may not have been clear from the flow diagram.

Operations. Production orders reach the shop floor through the dispatching function. Materials and components are issued by inventory control. Production is accomplished, products are returned to inventory control, and progress information is fed back through the data acquisition function.

Basic Research. Basic research originates ideas for new products essentially independently of market considerations. An example would be the development of nylon by the du Pont research group. This development opened the way for hundreds of commercial uses. However, many research ideas prove fruitless.

Product Innovation. Ideas for new products and modifications of existing products are consolidated by this function, whether the ideas originate in product markets, basic research, an independent inventor, or a member of the product engineering group.

Engineering. Specific configurations of all models in the product line are finalized here, often in consultation with product innovation. The manufacturing engineering functions, including the determination of manufacturing operations, raw material and purchased parts requirements, standard processing times, setup times, and so on, are also performed here. Standard operating procedures for performing the operations are prepared. The complete file for each new or modified product is conveyed to operations planning. Important feedback, including cost information from the production process, quality control information, labor usage, material usage, scrap rates, and so on, comes from measures of effectiveness.

Purchasing. Purchase orders are received from inventory control. A vendor is selected, and the order is placed. Material costs are conveyed to financial control.

Financial Control. This function determines and reports the effects of the company's activities upon its financial position. Standard costs are developed from production records, material costs, line labor, and overhead rates. Labor charges are

accumulated against work orders, accounts, and departments. Reports are produced which cover variances between actual and standard costs of material, labor, and overhead. Budget constraints are transmitted to operations planning. Vendors are paid for raw materials. Production costs are received from operations, and sales dollars are received from sales.

7.2.7 Integrating the Functions

There is, of course, considerable overlap among the five functions discussed in preceding sections. The performance of certain of these functions depends on the previous performance of other functions, which may depend on still others. It is usually necessary, therefore, to perform some functions essentially simultaneously, while performing certain groups of functions iteratively. The concept of an integrated operations planning and control system is explored further in Section 7.8.

7.3. Techniques for Demand Forecasting

Demand forecasting provides the major input to the other functions in the operations planning and control system. The other functions convert this forecast into material requirements, parts lists, labor requirements, schedules, and other decisions.

All planning must begin with an estimate, or forecast, of the amount of business our company can expect during the planning period. The means by which the estimate is arrived at may be completely subjective or unscientific, but the fact remains that all other planning of the company's activities depends on an estimate of business volume.

In this section we examine some of the elementary techniques of demand forecasting.

7.3.1 Moving Average

A commonly used forecasting technique is the *moving average method*. This method generates next period's forecast by averaging the actual demand values for the last n time periods. Mathematically, we have

$$\hat{x}_t = \frac{\sum_{i=1}^n x_{t-i}}{n} \quad (7.1)$$

where $\hat{x}_t$ = forecasted demand for period t

x_{t-i} = actual demand for i th period preceding t

n = number of time periods to include
in the moving average

The choice of the value for n is arbitrary and should be determined by experimentation (perhaps in a computer simulation model).

A variation of the procedure above is the *weighted moving average*, in which more weight is given to more recent data. The forecasting equation is

$$\hat{x}_t = \frac{\sum_{i=1}^n c_i x_{t-i}}{n} \quad (7.2)$$

where c_i = weight given to i th actual value

$$\sum_{i=1}^n c_i = n$$

The choice of values for n and the weighting coefficients are arbitrary and can be determined by experimenting with several combinations.

7.3.2 Exponentially Weighted Moving Average

The exponentially weighted moving average (EWMA) is a forecasting method which overcomes many of the disadvantages of the ordinary moving average and weighted moving average methods. Like the weighted moving average method, the EWMA method gives more weight to newer data; however, the EWMA method is much more efficient and is more readily adapted to computer application.

EWMA assigns weights to the demand values of previous periods in inverse proportion to their age. It does this in a very ingenious manner in which only three pieces of data are required to generate next period's forecast: (1) last period's forecast, (2) last period's actual demand, and (3) a smoothing constant, which determines the relative amount of weight given to recent demand values.

The rationale behind EWMA is that we would like our forecasting method to track the demand pattern through its general trends (i.e., its "ups and downs") without overreacting to purely random fluctuations. Suppose, for example, that for a particular product we forecasted a demand of 100 units. When actual demand became known, it was only 95 units. We must now forecast demand for the following month. Our question now becomes: How much of the difference between 100 units and 95 units can be attributed to an actual shift in the demand pattern and how much can be attributed to purely random causes? If our next forecast is for 100 units, we would be assuming that all of the difference should be attributed to chance and, therefore, that the demand pattern has not shifted at all. If our next forecast is for 95 units, we would be assuming that the entire difference should be attributed to a shift in the demand pattern and none to chance causes.

EWMA attributes part of the difference between actual demand and the forecasted demand to a trend shift and the remainder to chance causes. In our example, suppose we decide that 20% of the difference should be attributed to a trend shift and the remaining 80% to chance causes. Our new forecast, then, would be determined by reducing the previous forecast by 20% of the observed difference of 5 units. Since our previous forecast was 100 units and 20% of 5 units is 1 unit, the new forecast would be 99 units. After the actual demand for the new period became

known, we would determine the forecast for the following period by again computing 20% of actual demand minus forecasted demand and adding that quantity (which may be positive or negative) to the previous forecast, or 99. The 20% value is called the *smoothing constant* and is denoted α .

We can generalize this procedure as follows:

$$\begin{array}{lcl} \text{Next} & = & \text{previous} + \text{smoothing} \times \text{forecast} \\ \text{forecast} & & \text{forecast} \quad \text{constant} \quad \text{error} \\ \hat{x}_t & = & \hat{x}_{t-1} + \alpha(x_{t-1} - \hat{x}_{t-1}) \quad (0 \leq \alpha \leq 1) \end{array} \quad (7.3)$$

This expression can be rearranged algebraically to obtain an equivalent expression more convenient for computation.

$$\hat{x}_t = \alpha x_{t-1} + (1 - \alpha)\hat{x}_{t-1} \quad (7.4)$$

7.3.3 Regression Analysis

When a demand pattern is consistently increasing or decreasing, we can develop our forecasting transfer function by using regression analysis, the details of which may be found in almost all standard statistics texts.^{3,4}

The rationale for using regression analysis for forecasting is that the total set of factors that generated the cause system in the past will continue to operate in the future. Thus, forecasting becomes a matter of determining the general trend line and extrapolating this line into the future.

Returning to our hypothetical company, Gadgets, Inc., suppose that during a 12-month period a particular model of our product, super-gadget, had sales volumes as shown in Figure 7.4. These values are plotted (months 1 through 12) in Figure 7.5. The solid line plotted through the 12 points represents the general trend of sales. The dashed line extended from the solid line represents our best guess as to what the future demand will be.

There are two ways of determining the trend line. Often, as in this case, a good trend line can be fitted to a set of points "by eyeball." If more precision is desired, we can use regression analysis. This method fits a line to a set of observations such that the sum of the squared vertical distances of the observations from the line is minimized.

The basic equation for a straight line that expresses demand (x) as a function of time (t) is

$$\hat{x}_t = a + bt \quad (7.5)$$

where a is the intersection of the line with the vertical axis when $t = 0$ and b is the slope of the line. Our problem is to determine the values of a and b such that the

³Irwin Miller and J.E. Freund, *Probability and Statistics for Engineers*, 3rd ed. (Englewood Cliffs, N.J.: Prentice Hall, 1985).

⁴W.W. Hines and Douglas C. Montgomery, *Probability and Statistics in Engineering and Management Science*, 2nd ed. (New York: John Wiley & Sons, Inc., 1980).

Month	Demand
Jan,	500
Feb.	510
Mar.	480
Apr.	600
May	600
June	660
July	590
Aug.	700
Sept.	680
Oct.	740
Nov.	790
Dec.	760

Fig. 7.4. One-year sales records for super-gadget.

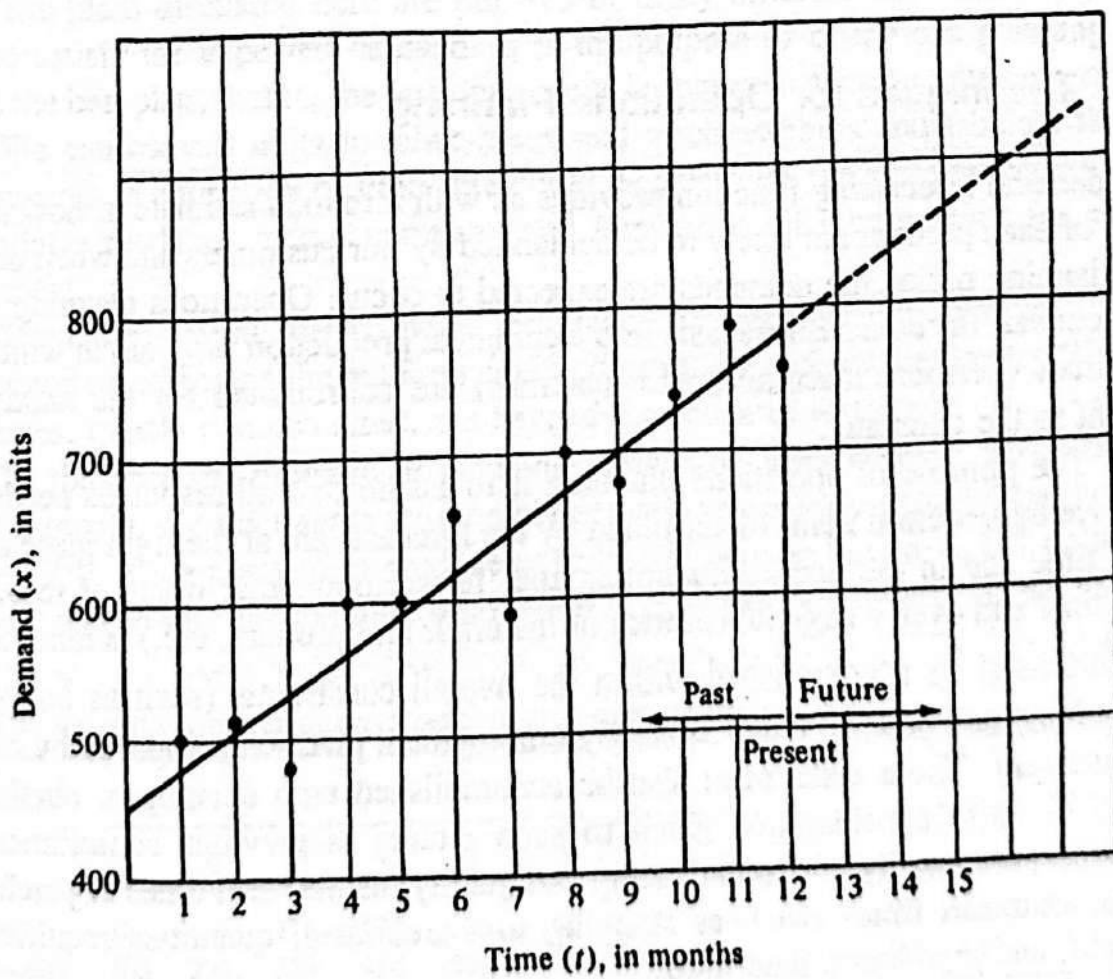


Fig. 7.5. Illustration of regression analysis in an upward trend.

least squares criterion is satisfied. The following equations for a and b are derived in almost all statistics texts:

$$b = \frac{n \sum_{i=1}^n t_i x_i - \sum_{i=1}^n t_i \sum_{i=1}^n x_i}{n \sum_{i=1}^n t_i^2 - \left(\sum_{i=1}^n t_i \right)^2}$$

$$a = \bar{x} - b\bar{t} \quad (7.6)$$

where n is the number of periods of demand data that we include in the calculations. Applying these equations to the data in Figure 7.4, we obtain

$$b = 27.8, \quad a = 452.6$$

Thus, our forecasting equation is

$$\hat{x}_t = 452.6 + 27.8t$$

which says that demand is increasing by an average of 27.8 units per month. Although we realize that our extrapolated line will not agree precisely with future demand values, it is expected that deviations of actual demand from the extrapolated line will be the result of random fluctuations from the line. Remember, it is the trend that we hope to predict accurately, not each individual demand value.

7.4. Techniques for Operations Planning

The demand forecasting function provides us with a refined estimate of how many units of each product are likely to be demanded by our customers and when during the planning period the demands are expected to occur. Operations planning must now convert the demand forecasts into a complete production program in which all resources (persons, machines, and materials) are coordinated for the maximum benefit to the company.

The purpose of operations planning is to assure that all resources needed to produce the required items (determined by the forecast) are at the right place at the right time and in the needed quantities and, furthermore, that waste of resources (idle time and overly large inventories of materials and product, etc.) is minimized. All this must be accomplished within the overall constraints (such as budgetary limitations) and policies (such as steady employment practices) imposed by higher management. These tasks must also be accomplished with existing or obtainable resources, with consideration given to such factors as previous commitment of resources, seasonality of demand, scrap and quality factors, lead times of purchased items, in-transit times (such as shipping to warehouse), quantities required in-process, and production lead times.

The operations plan provides only the general framework within which specific

activities are to be performed. The plan contains reasonable flexibility, so that detailed task assignments can be made within the general framework.

It is helpful in operations planning to represent various requirements as a function of time. Suppose that a certain super-gadget model has a seasonal sales pattern and that we have forecasted its demand for the next 12 months to be the values shown in Table 7.1. The cumulative demands are also shown.

The cumulative demand values are plotted as a solid line in Figure 7.6.

The significance of the cumulative demand curve is that through a combination of inventory and production, we must have available at least the cumulative values by each of the corresponding time periods. At any one time our combined inventory and production can exceed the cumulative demand, but it cannot be less (if no backordering is permitted).

Two possible production plans (production plus inventory) are plotted as dashed lines. Plan 1 permits a level production rate of about 70 units per month, but assumes that we begin the year with about 340 units in inventory. Note also that we end the year with almost no inventory. This condition could be undesirable for next year's planning. Plan 2 employs a production rate of 120 units per month for the first 6 months and 60 units per month for the last 6 months. In plan 2 we must start the year with only 100 units in inventory, but this plan also leaves us with almost no inventory at the end of the year. The two plans are summarized in Table 7.2.

The plans discussed here are but two of many different ones that could be used to satisfy the expected demand. It is the purpose of operations planning to devise the best plan, that is, the one that results in lowest total cost to the company.

We can convert units to labor-hours and machine-hours and tabulate these requirements as a function of time. We could then perform the same procedure for all products produced in our company and summarize all requirements for each department.

There are several matrix-based techniques for generating requirement-time profiles and summarizing the requirements by department and machine center. These techniques, mostly computerized, are beyond the scope of this book.

To illustrate the problem of planning production when demand varies from month to month, consider again the monthly demand values for super-gadgets shown in Table 7.1. If we planned to produce exactly 103 units in month 1, 117 units in month 2, and so on, our monthly production rate would vary greatly. If we had a

Table 7.1. MONTHLY AND CUMULATIVE DEMAND FORECASTS IN UNITS

	Month											
	1	2	3	4	5	6	7	8	9	10	11	12
Demand	103	117	115	121	123	109	89	74	71	73	81	98
Cumulative	103	220	335	456	579	688	777	851	922	995	1,076	1,174

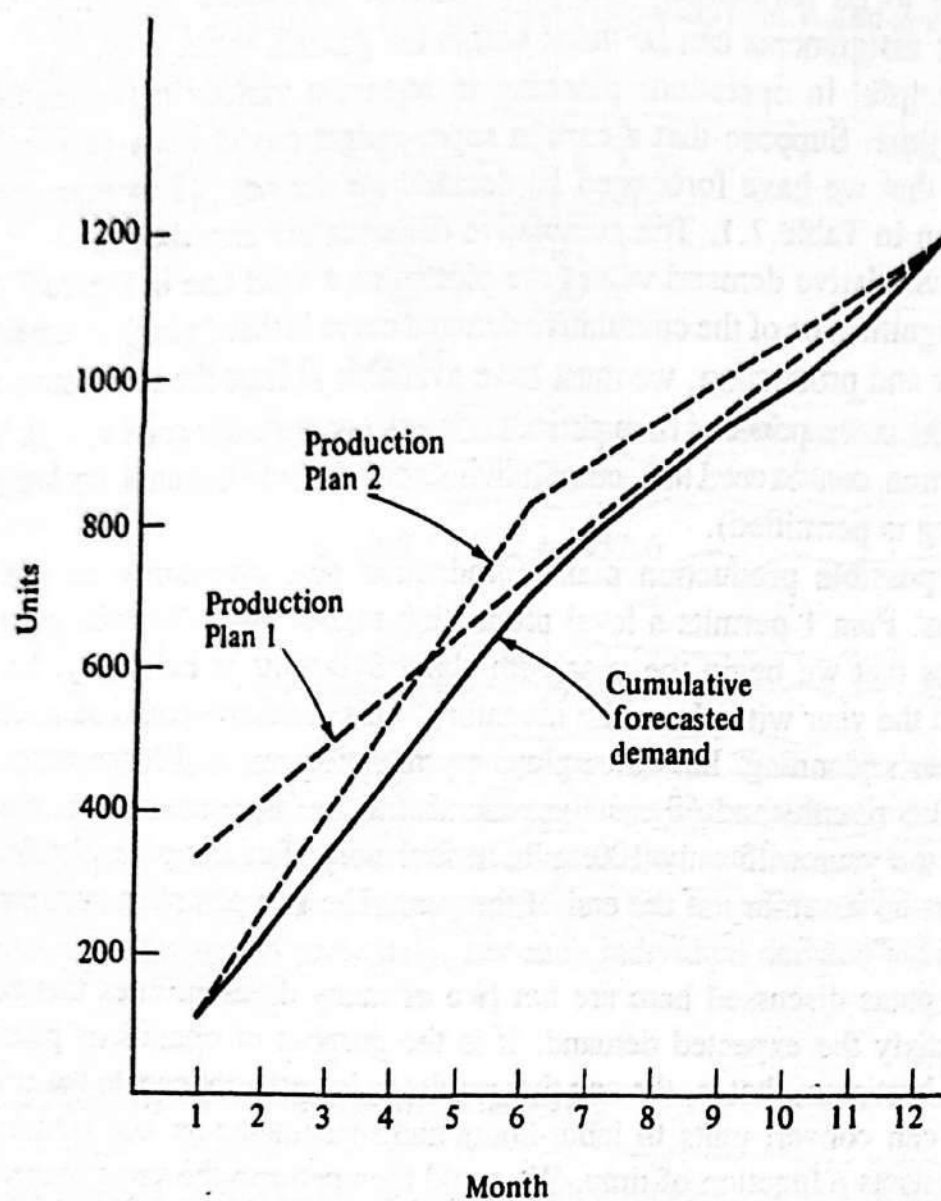


Fig. 7.6. Cumulative demand forecast for Table 7.1.

Table 7.2. PRODUCTION PLANS 1 AND 2 OF FIGURE 7.6

Month	Demand Forecast	Cumulative Forecast	Production Plan 1			Production Plan 2		
			Begin Inventory	Production	End Inventory	Begin Inventory	Production	End Inventory
1	103	103	340	70	307	100	120	117
2	117	220	307	70	260	117	120	120
3	115	335	260	70	215	120	120	125
4	121	456	215	70	164	125	120	124
5	123	579	164	70	111	124	120	121
6	109	688	111	70	72	121	120	132
7	89	777	72	70	53	132	60	103
8	74	851	53	70	49	103	60	89
9	71	922	49	70	48	89	60	78
10	73	995	48	70	45	78	60	65
11	81	1,076	45	70	34	65	60	44
12	98	1,174	34	70	6	44	60	6

constant size work force, we would be forced to work a lot of overtime during the high-demand months, and there would be a lot of idle time during the low-demand months.

If, however, we insisted on a level production rate, we would either have to lose sales during the high-demand months or carry a high inventory of finished goods to meet demand during those periods.

There are extra costs associated with both of the policies described above. The first cost involves overtime and idle time. The second cost involves inventory carrying costs.

Our objective is to smooth production over the planning period (usually one year) such that total cost is minimized.

Fairly good results can be achieved in smoothing production over the year simply by "eyeballing" it. More elaborate procedures are presented in most modern texts on production planning and control.

7.5. Techniques for Inventory Planning and Control

The demand forecasting function (Section 7.3) provides us with an estimate of demand for each of our products. Forecasts are generated for each operating period, and these values are summed to determine cumulative forecasts for the entire planning period. The operations planning function (Section 7.4) converts the forecasts into a time-phased production program. Units of finished product are translated into material requirements of raw materials, purchased parts, fabricated parts, and assemblies. Inventory planning and control must now provide procedures that, while assuring that materials and products will be available when they are needed and in the quantities required, will also guard against the costs of excessive inventories. The purpose of the inventory planning and control function is to determine appropriate inventory policies and to keep all associated costs at a minimum.

Almost all inventory problems are concerned with answering two fundamental questions:

- (1) *How much* to order (either from a supplier or from our own production facilities) at one time?
- (2) *When* (or *how often*) to place an order?

There are opposing costs in each of these questions. In the first question there is a cost associated with ordering too much at one time and another associated with ordering too little at one time. In the second question there is a cost associated with placing orders too frequently and another associated with not placing orders frequently enough. Our general objective is to determine the course of action that minimizes the sum of all such costs.

The expression for total costs depends on the particular situation. Several models are developed later in this section. It would be instructive at this point, however, to identify the major classes of inventory costs.

- **Procurement Cost**—The procurement cost of a purchased item is the order cost and consists of the clerical cost of making up and processing a purchase requisition. In general, order cost includes any cost whose size or amount is affected by the number of orders processed during a given time period.
- **Carrying Cost**—Carrying cost is the cost of holding inventory and includes several component costs:
 - (1) The cost of money invested in inventory which could be used in other ways.
 - (2) The cost of storage space—warehousing costs, utilities such as heat and light, etc.
 - (3) The cost of obsolescence, spoilage, and pilferage.
 - (4) The cost of insurance and taxes on the items in inventory.

The classical inventory model determines the optimal number of units to order on each order placed. This number is derived under the following assumptions:

- (1) The usage rate of the item is linear and known with certainty.
- (2) The order is received instantaneously when placed (i.e., lead time is zero).

An inventory system operating under these idealized conditions would behave as shown in Figure 7.7. Under the conditions we have assumed, we can wait until the inventory level reaches zero, then place an order for the desired amount. This order is received immediately. Our objective is to determine the number that should be ordered at one time.

We can express the total cost as the sum of procurement cost and carrying cost:

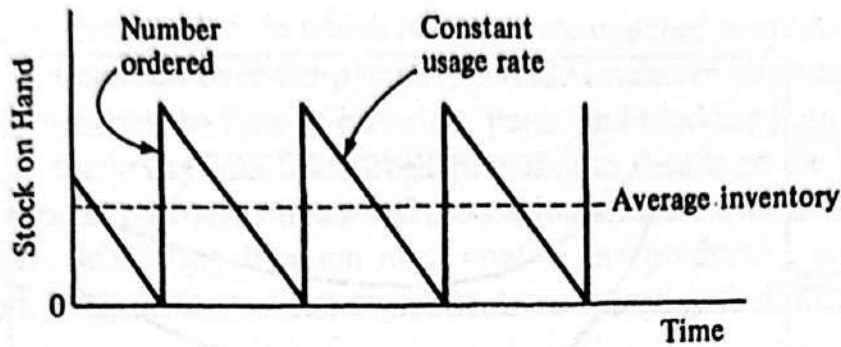
$$\text{total cost} = \text{procurement cost} + \text{carrying cost}$$

We use the following symbols in the development of the model:

- TC = total cost
- PC = procurement cost for each order
- CC = carrying cost per unit per year
- D = annual demand (or usage) of the item
- Q = quantity ordered on each order placed (lot size)
- Q_0 = optimal lot size

To determine the particular lot size that will result in the lowest value for total cost, we develop a total cost expression in which the two opposing costs, procurement cost and carrying cost, are in terms of Q , the quantity ordered.

We can express the procurement cost in terms of Q by recognizing that the total annual procurement is equal to PC (the procurement cost for each order) multiplied by the number of orders placed during the year. (The number of orders placed during the year is just the total annual demand divided by the number of units ordered on each order.) Symbolically, therefore, we have



* Fig. 7.7. Idealized inventory system.

$$\text{annual procurement cost} = PC \frac{D}{Q} \quad \left\{ \begin{array}{l} \# \text{ of orders} \\ \text{placed} \end{array} \right.$$

We can express the carrying cost in terms of Q by recognizing that the total annual carrying cost is equal to CC (carrying cost per unit per year) multiplied by the average number of units held in inventory. It is clear from Figure 7.7 that the average number of units held in inventory is just $Q/2$. Therefore,

$$\text{annual carrying cost} = CC \frac{Q}{2}$$

Our total cost equation can now be written as

$$TC = PC \frac{D}{Q} + CC \frac{Q}{2} \quad (7.7)$$

For a particular inventory problem we will know values for PC , CC , and D . With these values fixed, total cost becomes a function only of Q , the lot size. Suppose, for example, that $PC = \$10.00$ per order, $CC = \$0.20$ per unit per year, and $D = 10,000$ units. If we order 2,000 units at a time, then from Equation (7.7) we obtain

$$\begin{aligned} TC &= \$10.00 \frac{10,000}{2,000} + \$0.20 \frac{2,000}{2} \\ &= \$50 + \$200 = \$250 \end{aligned}$$

Other total cost values would be obtained by using different lot sizes. In fact, it is instructive to plot several values of Q , as has been done in Figure 7.8, to show its effect on each cost component in Equation (7.7), as well as on TC itself.

Our objective is to determine the particular value of Q that will minimize our total cost equation. In Figure 7.8 we see that the minimum value on the TC curve occurs at a point where the curve has zero slope. We know from differential calculus that the first derivative of a mathematical function gives the slope (instantaneous rate of change) of the function at a particular point. In our case, we are hunting the point where the slope is zero. Our approach, then, is to take the first derivative of

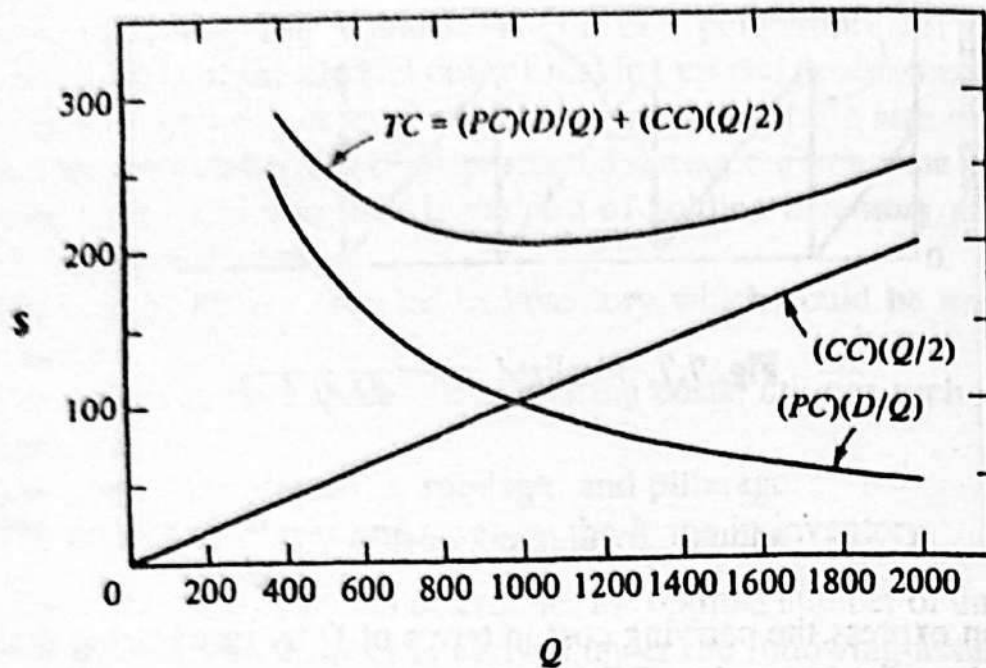


Fig. 7.8. Total cost function of classical inventory model where $D = 10,000$ units per year, $CC = \$0.20$ per unit per year, and $PC = \$10,000$ per order.

the total cost equation, Equation (7.7), set the derivative equal to zero, and then solve for Q_0 , the optimal lot size:

$$\frac{d(TC)}{dQ} = -PC \frac{D}{Q^2} + \frac{CC}{2} \quad (7.8)$$

$$-PC \frac{D}{Q^2} + \frac{CC}{2} = 0 \quad (7.9)$$

$$Q_0 = \sqrt{\frac{2(PC)(D)}{CC}} \quad (7.10)$$

By substituting our example values into Equation (7.10), we obtain

$$Q_0 = \sqrt{\frac{2(\$10.00)(10,000)}{\$0.20}} = 1,000 \text{ units}$$

We see from Figure 7.7 that a lot size of 1,000 units does indeed result in lowest total cost of \$200.

Several other models are presented in other texts which allow for such factors as variable demand, lead time, and safety stocks.

7.6. Techniques for Operations Scheduling

Demand forecasting (Section 7.3) provides an estimate of demand for each product during each operating time period. Operations planning (Section 7.4) develops a

time-phased production plan, in which resources are matched to production requirements and are smoothed over the planning period. Inventory planning and control (Section 7.5) regulates the flow of materials, parts, and products into, through, and out of the production system. This regulatory action results in the placement of purchase orders for purchased items and production orders for manufactured items. The operations scheduling function must now fit the production orders into the existing work loads of the various departments or operating facilities.

7.6.1 Purpose of Operations Scheduling

→ *Scheduling theory*

The purpose of operations scheduling is to assign specific operations to specific operating facilities with specific start and end times indicated.

Operations scheduling attempts to assign work to the required facilities in such a way that all the various costs associated with manufacturing are minimized. These costs are caused by factors such as in-process inventory, idle workers and equipment (for whatever reasons), overtime, orders completed behind schedule, and so on.

To fully appreciate the magnitude and importance of the scheduling function, we must first attempt to visualize the complexity of large manufacturing operations. Orders (regular production orders from inventory control and special orders through sales) are received periodically. Individual jobs are assigned to specific facilities. Certain precedence restrictions must be observed. As jobs progress through the facilities, they often must compete with other jobs for the same scarce resources. This competition results in conflicts on various facilities. Machines break down, some workers fail to appear and others perform below or above standard, tools break or wear out, materials are defective, and machines go idle waiting for work from preceding work centers. Orders are canceled, reduced, or increased. Raw materials fail to arrive when expected. Sales drop suddenly or increase sharply. Marketing begins a crash advertising campaign that causes a sudden increase in demand, after which demand will oscillate sharply until the system is allowed to stabilize again. Engineering introduces product design modifications that alter standard processing times, setup times, operation sequences, operator instructions, etc.

It is very difficult to exercise close control over a dynamic environment such as that described above. In spite of the difficulties, however, the scheduling function must be performed. The objectives of operations scheduling are:

- (1) High percentage of orders completed on time.
- (2) High utilization of facilities and workers.
- (3) Low in-process inventory.
- (4) Low overtime.
- (5) Low stockouts of manufactured items.

These objectives must be accomplished within the overall framework specified by the operations plan.

One of the oldest techniques available for sequencing and scheduling opera-

tions to facilities is the *Gantt chart*. The Gantt chart is very useful in showing planned work activities versus actual accomplishments on the same time scale. There are numerous possible variations of the Gantt chart. Many are color coded to indicate certain conditions such as material shortages, machine breakdowns, and so forth.

The basic Gantt scheduling chart is shown in Figure 7.9. Operations assigned to each facility are sequenced into a feasible schedule. Actual progress is indicated on the same chart to show any deviations from planned operation process times. The status of the facilities can be read from the chart. Operations 44A and 65D have been completed. Operation 65E is exactly on schedule as of now. Operation 44C is a half-day behind schedule. Operation 86B is 2 days ahead of schedule. Operations 86D and 103C are scheduled but have not yet begun. We should recognize that the time scale may be in any appropriate time units—hours, days, weeks, or months.

When the Gantt chart is used to sequence several operations resulting from new production orders, a good rule to follow is to sequence the most heavily loaded facility first, the second most heavily loaded facility second, and so on.

Although a fairly good schedule can be generated in this way, it will not be optimal under any criterion except by accident.

The Gantt scheduling chart must be updated very frequently in order to reflect the changing status of the shop. Certain jobs will take longer than expected, while others will be completed ahead of schedule. Machines will break down, workers will be absent, and materials will run out. All these factors make it difficult to keep

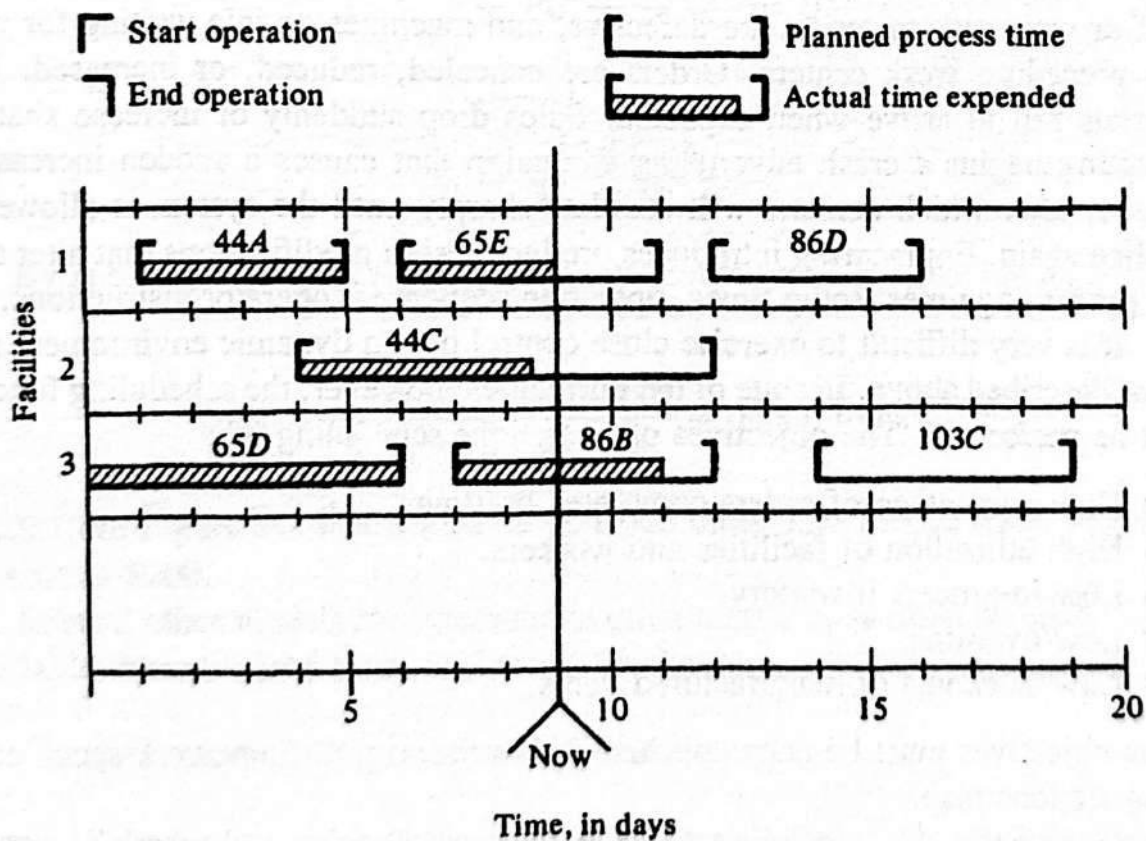


Fig. 7.9. Gantt scheduling chart.

the Gantt chart updated on a timely basis. Nevertheless, the Gantt chart should not be overlooked as an aid for loading, sequencing, and scheduling.

7.7. Dispatching and Progress Control

In the preceding section we concentrated primarily on the way decisions are made in an operations planning and control system. For example, we talked about inventory decisions, forecasting decisions, scheduling decisions, and the like. Now we must consider how all these decisions are implemented, that is, how they are transferred into action. Somehow, the decisions must be relayed from the production control office to the supervisors and workers on the shop floor; material has to be issued; instructions have to be gotten to workers; parts must move from one operation to the next. As work is accomplished, we must have a means of recording progress, comparing it to planned progress, and then taking whatever corrective action that seems appropriate.

We could say that the primary purpose of dispatching and progress control is to close the loop in our operations planning and control system. To accomplish this, we must provide ways to detect when a deviation has occurred, analyze what has happened, decide what corrective action would be appropriate, implement the corrective action by incorporating it into subsequent scheduling decisions, and finally, feed back progress information to the planning section to improve future planning.

Procedures for accomplishing this function are beyond the scope of this text. An essential component of such procedures is obviously a good information system. Consequently, this function is often performed with the aid of modern computer systems. See Chapter 20 for more details.

7.8. MRP Systems

One of the most important recent developments in operations planning and control is that of material requirements planning (MRP) systems, and their derivatives, MRPII (manufacturing resource planning) and closed-loop MRP. Details of MRP may be found in other sources.⁵ Our purpose here is to introduce the concepts briefly and illustrate MRP's usefulness in effectively managing an operation.

The basic concepts of operations planning were introduced in Section 7.4. The primary output of operations planning is a *master production schedule*, which is a listing of the specific products to be produced and the number of units of each product that will be produced in each time period. This becomes a primary input to MRP.

⁵T. E. Vollman, W. L. Berry, and D. C. Whybark, *Manufacturing Planning and Control Systems*, 2nd ed. (Homewood, Ill.: Richard D. Irwin, Inc., 1988).

Another primary input to MRP is the *bill of materials* (BOM) for each product. The BOM is a detailed breakdown of the various subassemblies, components, and raw materials that make up the product. Typically, this detailed breakdown is shown in the form of a *product structure diagram*, an example of which is shown in Figure 7.10. P1 indicates product 1; SA1 and SA2 indicate subassemblies 1 and 2; C1 through C5 indicate components 1, 2, 3, 4, and 5. The number in parentheses to the right of each box indicates the number of units of that item required to produce 1 unit of the item it goes into (i.e., the item at the next-higher level in the tree). For example, 4 C1s are required to produce 1 unit of SA1, and two SA1s are required to produce 1 unit of P1. Thus a total of 8 units of C1 is required to produce 1 unit of P1. The number just above each box indicates the lead time for that item. Lead time is the number of time periods required to obtain an order of a particular item. The lead time for SA2, for example, is 2 time periods.

The MRP program (usually, a computer software package) calculates the number of each subassembly and component required to produce a specified number of end products. It does this by “exploding” the end product requirements into successively lower levels within the product structure.

Suppose that our master production schedule specifies 200 units of product P1 needed in week 17. Referring to the product structure diagram in Figure 7.10, we can calculate the number of units of each subassembly and each component that will be needed:

SA1	400
SA2	200
C1	1,600
C2	800
C3	400
C4	200
C5	3,200

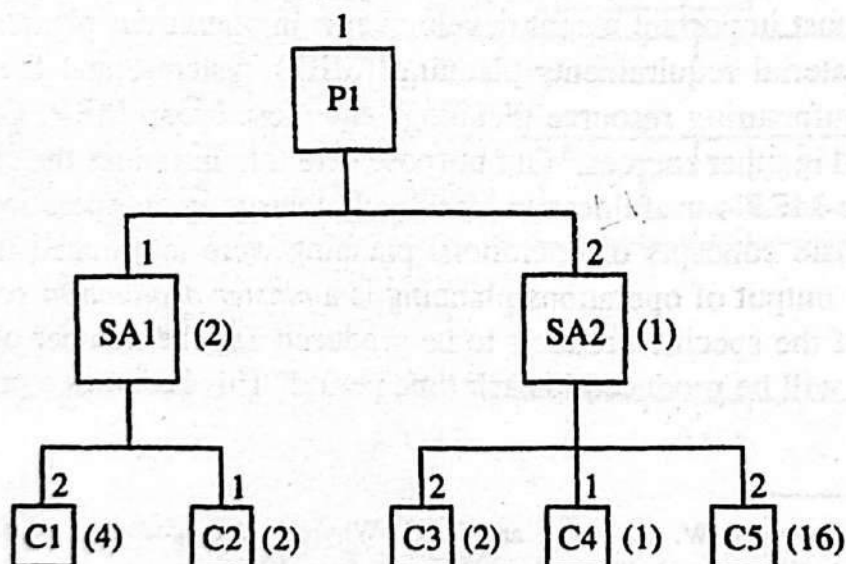


Fig. 7.10. Product structure diagram for product P1.

We must now consider the lead times required for each item. P1 has a lead time of 1 week. (For the remainder of this example, the time period will be considered as one week.) This means that 400 units of SA1 and 200 units of SA2 must be available at the end of week 16. The lead time for SA2 is 2 weeks. This means that 400 units of C3, 200 units of C4, and 3,200 units of C5 must be available at the end of week 14. Therefore, an order for 3,200 units of C5 must be placed no later than the end of week 12.

Notice that the calculated requirements for each item is the gross number of units needed in order to satisfy the master production schedule. If some of these items are already in inventory, or if an order is scheduled for delivery in the near future, these must be subtracted from gross requirements to determine net requirements for satisfying the master production schedule. For example, suppose that we already have 716 units of C5 in inventory. The net requirement for C5 is, therefore, $3,200 - 716 = 2,484$ units. Suppose further that C4 is a purchased item and that 75 units of C4 are scheduled to arrive in week 12. Our gross requirements for C4 were calculated earlier as 200 units to be available at the end of week 14. So our net requirements for C4 are $200 - 75 = 125$ units.

A systematic means of performing these calculations is needed. Numerous computer software packages are available for performing MRP calculations. For manual calculations, a matrix approach is convenient. For the calculations discussed above, the MRP solution is shown in Figure 7.11. Notice that only those calculations pertaining to the right branch (SA2) in Figure 7.10 are shown. Similar calculations would be required to determine the time-phased requirements for all items in the left branch (SA1).

Figure 7.11 is merely a "slice" of a total MRP solution, in that it illustrates the backward explosion calculations from the demand for P1 in week 17. Similar calculations would be made for the demand for P1 in all other weeks. These demand values, shown as "gross requirements" on the form, come from the master production schedule for end products. However, the "gross requirements" value for all subassemblies and components must be calculated according to the information provided on the product structure diagram (Figure 7.10).

When implemented properly into a manufacturing operation, a MRP system provides tight, accurate control over the many complex, interdependent activities of the plant. All activities are synchronized such that parts are being manufactured (or received from vendors, if purchased) exactly when they are needed for the next stage of production. Such a system avoids the typical problems of excess in-process inventory, unavailability of components when needed, grossly uneven utilization of resources, and so on.

When these concepts are extended to the entire range of manufacturing operations, and when corrective action is incorporated into the system, the result is a "closed-loop MRP system." In a closed-loop MRP system, the various functions of operations planning and control (forecasting, operations planning, inventory management, MRP calculations, dispatching, and progress control) have been integrated into one unified system. It also provides for feedback from vendors

Period	10	11	12	13	14	15	16	17	18	19
Item: P1 LT 1										
Gross Requirements								200		
Scheduled Receipts										
On Hand 0										
Net Requirements								200		
Planned Order Releases							200			
Item: SA 2 LT 2										
Gross Requirements							200			
Scheduled Receipts										
On Hand 0										
Net Requirements							200			
Planned Order Releases					200					
Item: C3 LT 2										
Gross Requirements					400					
Scheduled Receipts										
On Hand 0										
Net Requirements					400					
Planned Order Releases			400							
Item: C4 LT 1										
Gross Requirements					200					
Scheduled Receipts			75							
On Hand 0			75	75	75					
Net Requirements					125					
Planned Order Releases				125						
Item: C5 LT 2										
Gross Requirements					3200					
Scheduled Receipts										
On Hand 716	716	716	716	716	716	716				
Net Requirements					2484					
Planned Order Releases			2484							
Item: LT										
Gross Requirements										
Scheduled Receipts										
On Hand										
Net Requirements										
Planned Order Release										

Fig. 7.11. MRP calculations. [Form adapted from M. P. Groover, *Automation, Production Systems, and Computer Integrated Manufacturing* (Englewood Cliffs, N.J.: Prentice Hall, 1987), Fig. 24.10, p. 739.]

(delayed order deliveries, etc.) and from customers (changes in orders, models, etc.).

One further extension to the MRP concept can be made by linking the closed-loop MRP system to the financial function of the organization. This type of structure, called "manufacturing resource planning," is typically indicated by the label MRPII. Commercially available MRPII software usually includes a simulation capability. The simulator allows the manager to explore various "what if" questions. For example, if a particular customer requests that its order for 500 units be delivered 3 weeks earlier than the original contract date, the simulator allows the manager to determine whether this request can be granted without an adverse effect on other customer orders.

7.9. Just-in-Time Manufacturing⁶

A very important recent development in operations planning and control is that of just-in-time (JIT) manufacturing. Sometimes referred to as "stockless production," JIT is more than just a method to reduce inventory. It is also concerned with the total organization of the production process so that defect-free components (both manufactured and purchased) are available for the next stage of production exactly when they are needed—not too late and not too early.

A company implementing the JIT philosophy must attend to several important details:

- *Smooth Material Flow*—Simplify material flow patterns. This sometimes requires a total rearrangement of the production lines. It also requires direct access to and from receiving and shipping docks. The goal is to have material flow uninterrupted from receipt, directly through each production stage, then to delivery. Anything that interrupts this flow is a target for investigation and elimination.
- *Reduction of Setup Times*—Prior to JIT, many discrete part manufacturers had machine setups that sometimes required several hours. This is intolerable in a JIT system. Dramatic reductions in setup times have been achieved by many companies, sometimes from 4 to 7 hours to 3 to 7 minutes. This allows batch sizes to be reduced to very small values, thereby allowing the company to be very flexible and responsive to changes in customer requirements.
- *Reduction of Vendor Lead Times*—Instead of receiving large shipments of purchased parts every 2 or 3 months, under JIT we want to receive parts "just in time" for the needed production operation. Companies sometimes have to negotiate long-term contracts with suppliers in order to achieve this feature.
- *Zero-Defect Components*—A JIT system cannot tolerate defective compo-

⁶See Chapter 7, W. W. Luggen, *Flexible Manufacturing Cells and Systems* (Englewood Cliffs, N.J.: Prentice Hall, 1991).

nents, either manufactured or purchased. For manufactured components, statistical process control techniques must be implemented to assure that the process is producing parts within tolerance at all times. For purchased components, vendors are required to guarantee that the components they are supplying have been produced on processes that are in statistical control. A company will usually have a vendor certification program to assure that this occurs.

- *Disciplined Shop Floor Control*—In traditional shop floor control systems, the emphasis was on high machine utilization, relatively long production runs to keep setup costs low, and on low labor idle time. Consequently, production orders were released to the production units with these factors in mind. Under JIT, these traditional performance measures are far outweighed by the desire to keep inventories very low and to maintain an “unclogged,” responsive operation. This requires that strict adherence to precise order release times be adhered to at all times. This means that machines and machine operators are sometimes idle. Many production managers who have spent most of their professional lives keeping machines busy and operators working are having a difficult time making the necessary mental adjustments required for a successful JIT operation. For those companies that have successfully implemented the JIT philosophy, the rewards have been significant.